Chapter 7 Sequences and Series

Expectations

By the end of this unit, student will:

1. demonstrate an understanding of recursive sequences, represent recursive sequences in a variety of ways, and make connections to Pascal's triangle;
2. demonstrate an understanding of the relationships involved in arithmetic and geometric sequences and series, and solve related problems;

Homework Log

Day	Date	Section	Assigned Work	Questions to Take Up	Extra Practice
1		7.1 Arithmetic Sequences	Text: p. 424 #1, 5, 8 – 11, 13ef, 15		p. 424 #2 – 4, 6, 7
2		7.2 Geometric Sequences	Text: p. 430 #1, 6, 9 – 14		p. 430 #2 – 5, 7, 8
3		7.4 Recursive Sequences	LP: p. 9 #1 – 6		Text: p. 439 #1 – 10
4		7.5 Arithmetic Series	Text: p. 452 #4, 5acdf, 6bef, 7ace, 8, 10 – 13, 15, 16		p. 452 #1 – 3, 5be, 6acd, 7bdf
5		7.6 Geometric Series	Text: p. 459 #3bdf, 4, 5bdef, 6bcef, 11, 16		Online worksheets
6		7.7 Pascal's Triangle and Binomial Expansion	Text: p. 466 #1, 4, 10		p. 466 #2, 3, 5
7		Review	Text: p. 468 #3, 4, 5, 7, 8, 10, 11, 14eoo, 15eoo, 16, 17, 18eoo, 20, 21, 22eoo, 23eoo		

***** eoo – every other one

7.1 Arithmetic Sequences

A **sequence** is a list of numbers arranged in an _______________.

The numbers in a sequence are called _____________.

The first term is referred to as t_1, the second as t_2, and so on.

A sequence that has a last term is a _____________ **sequence**. A sequence that has no last term and continues indefinitely is an _______________ **sequence**.

A **recursive formula** gives the beginning term or terms of a sequence and then a recursive equation that tells how t_n is related to one or more preceding terms.

The n^{th} term is referred to as t_n (or the **general term** of the sequence), where n is a natural number referring to the position of the term in the sequence.

Ex.1: Given the formula for the n^{th} term, determine the first **three** terms of each sequence.

a) $t_n = 3n^2 - 4$

b) $f(n) = \dfrac{n-2}{n+2}$

Arithmetic Sequences

In an **arithmetic sequence** the difference between consecutive terms is ____________.

This constant is the **common difference**, ***d***.

The first term, t_1, is denoted by the letter ***a***.

Ex.2: Find the next three terms of each arithmetic sequence.

a) 3, 7, 11, ...

b) 25, 18, 11, ...

c) $\frac{3}{4}, \frac{5}{4}, \frac{7}{4}, \ldots$

Recursive Formula of an Arithmetic Sequence

General Term of an Arithmetic Sequence

Ex. 3: Find the general term, t_n, and the recursive formula for each arithmetic sequence. Also, find the indicated term.

a) 12, 16, 20, ... ; t_{41}

b) -3, 2, 7, ... ; t_{22}

Ex. 4: Determine the number of terms in the finite arithmetic sequences:

a) 3, 15, 27, ..., 495

b) -10, -14, -18, -22, ..., -138

Ex. 5: Given $t_{12} = 52$ and $t_2 = 102$, find a and d, then write the formula for t_n.

Ex. 6: A music hall has 35 seats in the front row, 44 seats in the second, 53 seats in the third, and so on.

a) How many seats are in the 12th row?

b) If the last row has 215 seats, how many rows are there in the music hall?

7.2 Geometric Sequences

In a **geometric sequence**, the ________ of any term, except the first one, to the previous term is constant for all pairs of consecutive terms.

This constant or **common ratio** is denoted by ___.

The **first term**, t_1, is denoted by the letter ___.

Ex.1: State the common ratio, *r*, for each geometric sequence.
a) 4, 8, 16, 32,...

b) 100, 10, 1, 0.1,...

What would the recursive formula of a geometric sequence?

What would be the general term of a geometric sequence?

Note: The relationship between *n* and t_n of any geometric sequence is ______________________.

Ex. 2: For the geometric sequence given below, find the general term and the recursive formula. Then, find t_{10}.
4, 12, 36, 108, ...

Ex. 3: Find the number of terms in the geometric sequence given below.
7, -14, 28, ..., -3584

Ex. 4: For a geometric sequence, $t_3 = 15$ and $t_7 = 0.9375$. Find *a*, *r*, and t_n.

Ex. 5: A community organizes a phone tree in order to alert each family of emergencies. In the first stage, one person calls 5 families. In the second stage, each of the five families calls another 5 families, and so on. How many stages need to be reached before 19,530 families or more have been called?

7.4 Recursive Sequences

Explicit Formulas

Explicit formulas can be used to calculate any term in a sequence without knowing the previous term

For example, $t_n = 2^{n-1}$ can be used to determine the geometric sequence 1, 2, 4, 8, 16, . . .

Some sequences can only be created using a **recursive formula**.

It is sometimes more convenient to calculate a term in a sequence from one or more previous terms in the sequence. Formulas that can be used to do this are called ***recursion formulas***.

Recursion Formulas

Recursion formulas consist of at least two parts. The parts give the value(s) of the first term(s) in the sequence, and an equation that can be used to calculate each of the other terms from the term(s) before it. An example is the formula

$t_1 = 5$

$t_n = t_{n-1} + 2$

The first part of the formula shows that the first term is 5. The second part of the formula shows that each term after the first term is found by adding 2 to the previous term.

The sequence is: _____, _____, _____, _____, . . .

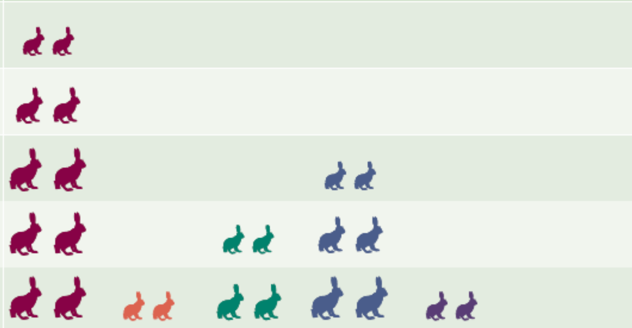
This recursion formula determines the same sequence as the explicit formula $t_n = 2n + 3$.

In one of his books, the great Italian mathematician Leonardo Fibonacci (c 1180 – c. 1250) described the following situation.

A pair of rabbis one month old is too young to produce more rabbits. But at the end of the second month, they produce a pair of rabbits, and a pair of rabbits every month after that. Each new pair of rabbits does the same thing, producing a pair of rabbits every month starting at the end of the second month.

1. The table shows how a family of rabbits grows. Extend the table to find the number of pairs at the start of the ninth month.

Start of Month	Number of Pairs	Diagram
1	1	
2	1	
3	2	
4	3	
5	5	



2. List the numbers of pairs of rabbits in order as the first 9 terms of a sequence.

______, ______, ______, ______, ______, ______, ______, ______, ______, . . .

This sequence is known as the ***Fibonacci sequence***. After the first two terms of the sequence, how can each term be calculated from previous terms?

3. Is the Fibonacci sequence arithmetic, geometric, or neither? Explain.

4. Is the Fibonacci sequence a function? Explain.

Key Ideas

- In a **recursive sequence**, a new term is generated from the previous term or terms. For example, 1, 2, −1, 3, −4, 7, −11, … is a recursive sequence because each term, beginning with the third term, is the result of subtracting t_{n-1} from t_{n-2}.
- A **recursive formula** shows how to find each term from the previous term or terms. For example,

  $t_1 = 1,\ t_2 = 2,\ t_n = t_{n-2} - t_{n-1}$, where $n \in N$

  To find t_n, subtract t_{n-1} from t_{n-2}. The first two terms are given.
  The sequence is 1, 2, −1, 3, −4, 7, −11, … .
- A recursive formula refers to at least one known term. The first term, and sometimes several other terms, appear with the formula. Examine the formula carefully before applying it.

Ex 1: Write the first four terms of each sequence.

a) $t_1 = 2,\ t_n = 3t_{n-1} + 5$

b) $t_1 = -1,\ t_2 = 1,\ t_n = 2t_{n-2} + 4t_{n-1}$

Ex 2: Write a recursive formula for each sequence.

a) 1, 2, 4, 7, 11, 16, …

b) 4, 5, 20, 100, 2000, …

c) 100, 96, 87, 71, 46, ….

Recursive Sequences Worksheet

1. Use the recursive formula to write the next 4 terms of each sequence.

a) $t_1 = 0, t_n = t_{n-1} - n^2$, where $n > 1$

b) $t_1 = 5, t_n = n^2 - t_{n-1}$, where $n > 1$

c) $t_1 = 3, t_n = \frac{t_{n-1}}{n}$, where $n > 1$

d) $t_1 = 2, t_n = (t_{n-1})^2 + 2$, where $n > 1$

e) $t_1 = 4, t_2 = 2, t_n = t_{n-1} - t_{n-2}$, where $n > 2$

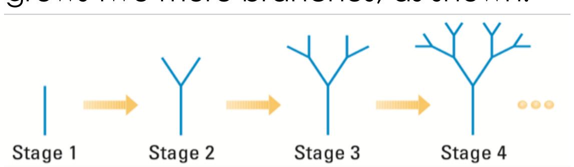
f) $t_1 = 2, t_n = n^2 + n - t_{n-1}$, where $n > 1$

2. Determine a recursive formula for each sequence. The sequence may be arithmetic, geometric or neither.

a) 5, 11, 17, 23, …
b) -4, -2, -1, -½, …
c) 1, 2, 2, 4, 8, 32, …
d) 2, 5, 26, 677, …
e) 3, 8, 63, 3968, …
f) 2, 5, 10, 50, 500, …
g) 1, 2, 3, 6, 11, 20, …
h) $6, 6\sqrt{2}, 12, 12\sqrt{2}, ...$
i) 3, 7, 16, 32, …
j) -1, 0, 3, 12, …

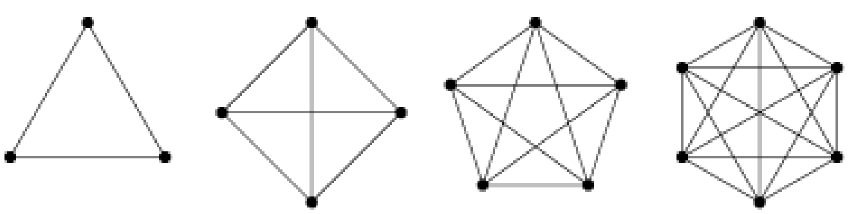
3. A lake initially contains 5200 fish. Each year the population declines 30% due to fishing and other causes, and the lake is restocked with 400 fish. Write a recursive formula for the number of fish at the beginning of the nth year. How many fish are in the lake at the beginning of the fifth year?

4. A fractal tree starts with a single branch (the trunk). At each stage the new branches from the previous stage each grows two more branches, as shown.



a) List the number of branches in each of the first seven stages.
b) Write a recursive formula for the number of branches in each stage.

5. The diagrams show the diagonals in regular polygons with n sides. Write the sequence for the number of diagonals and determine the recursive formula for this sequence.



6. A new theatre is being built for a youth orchestra. This theatre has 50 seats in the first row, 54 seats in the second row, 62 seats in the third row, 74 seats in the next row, and so on. Write a recursive formula to represent the number of seats in any row.

Answers

1. a) 0, -4, -13, -29, -54 b) 5, -1, 10, 6, 19 c) $3, \frac{3}{2}, \frac{1}{2}, \frac{1}{8}, \frac{1}{40}$

   d) 2, 6, 38, 1446, 2090918 e) 4, 2, -2, -4, -2, 2 f) 2, 4, 8, 12, 18

2. a) $t_1 = 5, t_n = t_{n-1} + 6$, where $n > 1$

   b) $t_1 = -4, t_n = \frac{1}{2} t_{n-1}$, where $n > 1$

   c) $t_1 = 1, t_2 = 2, t_n = t_{n-2} \times t_{n-1}$, where $n > 2$

   d) $t_1 = 2, t_n = (t_{n-1})^2 + 1$, where $n > 1$

   e) $t_1 = 3, t_n = (t_{n-1})^2 - 1$, where $n > 1$

   f) $t_1 = 2, t_2 = 5, t_n = t_{n-2} \times t_{n-1}$, where $n > 2$

   g) $t_1 = 1, t_2 = 2, t_3 = 3, t_n = t_{n-3} + t_{n-2} + t_{n-1}$, where $n > 3$

   h) $t_1 = 6, t_n = \sqrt{2} t_{n-1}$, where $n > 1$

   i) $t_1 = 3, t_n = n^2 + t_{n-1}$, where $n > 1$

   j) $t_1 = -1, t_n = t_{n-1} + 3^{n-2}$, where $n > 1$

3. $t_1 = 5200, t_n = 0.7 t_{n-1} + 400$, where $n > 1$, approximately 2262 fish

4. a) 1, 3, 7, 15, 31, 63, 127, … b) $t_1 = 1, t_n = t_{n-1} + 2^{n-1}$, where $n > 1$

5. 0, 2, 5, 9, …, $t_1 = 0, t_n = t_{n-1} + n$, where $n > 1$

6. 50, 54, 62, 74, … $t_1 = 50, t_n = t_{n-1} + 4(n-1)$, where $n > 1$

7.5 Arithmetic Series

Problem: What is the sum of the first 20 numbers?

What about the sum of the first 100 numbers?

Carl Friedrich Gauss (1777 – 1855), when only eight years old, used the following method to sum the natural numbers from 1 to 100.

A **series** is the sum of the terms in a sequence.

An **arithmetic series** is the sum of the terms in an arithmetic sequence.

We can use Gauss' method to derive a formula for the sum of the general arithmetic series:

Decide which one to use based on the information given.

Ex. 1 For the given arithmetic series, calculate S_{17}.

$3 + 7 + 11 + ...$

Ex. 2 Find the sum of the first 12 terms of the arithmetic series with $a = 3$ and $t_{12} = 36$.

Ex. 3 Find the sum of the first 25 terms of the arithmetic series where the 14^{th} term is 102 and the terms decrease by 9.

Ex. 4 In an amphitheatre, seats are arranged in 50 semicircular rows facing a domed stage. The first row contains 23 seats, and each row contains 4 more seats than the last. How many seats are there in total?

Ex. 5 Samantha deposited $128 into her bank account. Each week, she deposits $7 less than the previous week until she makes her last deposit of $9. Find the total value of her deposits.

7.6 Geometric Series

A **geometric series** is the sum of the terms in a geometric sequence.

For example, for the geometric sequence 7, 14, 28, 56, ... , the **geometric series** is

7 + 14 + 28 + 56 +... .

Where t_n represents the value of the n^{th} term, S_n represents the sum of the first n terms.

The sum of the first n terms of a geometric sequence (a series) can be calculated in two ways:

Ex. 1 For the given geometric series, calculate t_7 and S_7.

5 - 10 + 20 - ...

Ex. 2 Find the sum of the first 9 terms of the geometric series where the first term is -128 and the sixth is -4.

Ex. 3 Find the sum of the first 10 terms of the geometric series where the 11th term is 3 and the terms decrease by a factor of $\frac{1}{3}$.

Ex. 4 Calculate the sum of the geometric series.

$5 - 15 + 45 - \ldots + 3645$

Ex. 5 A catering company has 11 customer orders during its first month. For each month afterward, the company has double the number of orders than the previous month. How many orders in total did the company fill at the end of its first year?

Ex. 6 At a fish hatchery, fish hatch at different times even though the eggs were all fertilized at the same time. The number of fish that hatched were 2, 10, 50, 250, ..., 3906250. If the pattern continues, calculate the total number of fish hatched.

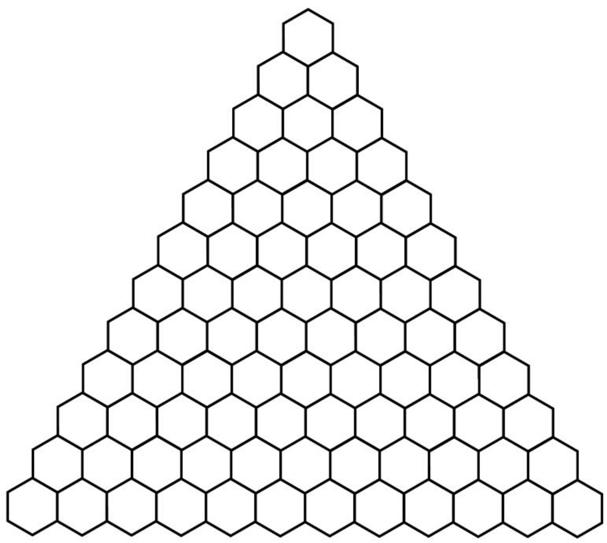
7.7 Pascal's Triangle and Binomial Expansion

What is Pascal's Triangle?

- A triangular arrangement of numbers.
- Name after French mathematician, Blaise Pascal (1623-1662), who brought the triangle to the attention of Western mathematicians.
- It was known as early as 1300 in China, where it was known as "Chinese Triangle".
- It is used to solve problems in algebra and probability.

How does it work?

Pascal's triangle can be built using a recursive procedure. Each term in Pascal's triangle is the sum of the two terms immediately above it. The first and last terms in each row are 1 since the only term immediately above them is always 1.



Among its many other applications, Pascal's Triangle is useful in performing binomial expansions.

Let's expand! (Look for a pattern)

$\left(x+y\right)^1$

$\left(x+y\right)^2$

$\left(x+y\right)^3$

$(x+y)^4$

$(x+y)^5$

Describe any patterns you see from the previous expansions.

Ex. 1. Expand.

a) $(m-4)^3$

b) $(2x^2+1)^4$

c) $\left(5x - 3y\right)^5$

Ex. 2. In a binomial expansion of $(a+2b)^n$, a term including a^3b^4 occurs.

a) What is the value of *n*?

b) What is the coefficient of a^3b^4?